

NIIS HACKATHON 2026 — OFFICIAL PROBLEM BRIEF

Track: Meditech AI

Challenge ID: PS-05

Title: AI-Based Heatwave Health Risk & Local Alert System

1. Executive Summary & Problem Overview

Extreme heat events and urban heat island effects are intensifying across India, causing severe public health crises, heat-induced illnesses (heat stroke, heat exhaustion, hyperthermia), and occupational hazards for outdoor laborers, delivery workers, elderly citizens, and children. Traditional meteorological bulletins provide broad city-wide temperature forecasts but fail to deliver actionable, localized physiological risk assessments. Conventional alerts often overlook critical micro-factors such as relative humidity, wet-bulb globe temperature (WBGT), solar radiation exposure, geographic altitude, and individual physiological vulnerability.

The objective of **PS-05 (AI-Based Heatwave Health Risk & Local Alert System)** is to develop a software-based, data-driven platform that integrates publicly available meteorological feeds, heat-index calculations, and geospatial demographic datasets. The system will predict localized heat hazards, stratify vulnerability scores, and generate automated, preventive, vernacular health advisories for high-risk demographics and municipal disaster authorities.

2. Core Personas & Access Roles

1. **Vulnerable Citizens & Field Workers:** Outdoor workers, street vendors, athletes, seniors, and parents who receive personalized hydration schedules, localized danger-window warnings, and nearby cooling center locators.
2. **Municipal Health & Disaster Officers:** Administrators who monitor real-time heat vulnerability heatmaps, coordinate emergency water stations, declare localized heat alerts, and manage hospital heat-stroke ward capacities.
3. **Primary Healthcare Workers (ASHA / ANM / PHC Staff):** Frontline personnel receiving early community risk flags to distribute oral rehydration salts (ORS) and conduct preventive wellness checks.
4. **Public Dashboard Viewers:** Community members who view live color-coded neighborhood heat-risk indices, UV alerts, and general first-aid guidance for heat emergencies.

3. Key Functional Modules & Technical Deliverables

Module 1: Weather Telemetry & Heat-Index Engine

- **Public Meteorological Data Ingestion:** Automates ingestion from free/public APIs (Open-Meteo, IMD, NASA POWER, NOAA, or visual weather APIs) for ambient temperature, relative humidity, wind speed, and solar radiation.
- **Bio-Meteorological Index Computation:** Calculates physiological indices including the Steadman Heat Index, Humidex, and estimated Wet-Bulb Globe Temperature (WBGT) to gauge actual thermal strain.
- **Micro-Climate & Urban Hotspot Mapping:** Analyzes land-use factors, vegetation density (NDVI approximations), and population density to identify localized neighborhood heat islands.

Module 2: AI-Powered Vulnerability Stratification

- **Demographic Vulnerability Modeling:** Multi-factor risk engine that evaluates age cohorts, pre-existing comorbidities (cardiovascular, respiratory, renal diseases), and occupational exposure levels.
- **Predictive Risk Classification:** Machine learning models that classify next 24-to-72-hour thermal stress levels into tiered threat categories (Low, Moderate, High, Severe, Extreme Danger).
- **Hospital Surge Forecasting:** Early predictive warnings estimating potential spikes in heat-induced emergency clinic admissions based on heatwave duration patterns.

Module 3: Preventive Alert & Advisory Dispatch

- **Multilingual Vernacular Advisories:** Generates automated health alerts in plain English, Odia, Hindi, and regional dialects via SMS, WhatsApp, Web Push, and audio speech synthesis.
- **Targeted Activity Guardrails:** Context-aware work/rest cycle recommendations for laborers and schools (e.g., *"Cease direct sunlight labor between 11:30 AM and 3:30 PM today"*).
- **Emergency Cooling & Water Locator:** Interactive map displaying nearest public drinking water points, shaded shelters, and PHCs stocked with IV fluids and cooling equipment.

Module 4: Administrative Disaster Dashboard

- **City & District Heat Vulnerability Map:** Interactive geospatial choropleth map highlighting ward-level thermal severity.
- **Automated Action Plan (HAP) Checklists:** Structured workflows aligned with national Disaster Management Authority (NDMA) guidelines for executing emergency protocols.
- **Exportable Health Reports:** Automated PDF/CSV analytical summaries of seasonal heat exposure trends, peak hazard hours, and high-risk demographic distributions.

4. Recommended Technology Stack

Layer, Suggested Technologies

Frontend Web, React.js (Vite) / Next.js / Tailwind CSS / Leaflet.js / Mapbox GL / Recharts

Mobile / PWA, "Flutter, React Native, or Progressive Web App (PWA) with Push Notifications"

Backend Architecture, Python (FastAPI / Flask) or Node.js (Express)

Database & Caching, PostgreSQL with PostGIS / Supabase / MongoDB + Redis for forecast caching

Data Science & ML, "Python (Pandas, Scikit-learn, XGBoost, Prophet for time-series temperature trends)"

Public Data Sources, "Open-Meteo API, IMD datasets, Copernicus Climate Data, NASA POWER API"

Notifications & Vernacular, "Twilio API / WhatsApp Business API / Web Push API, Bhashini or Google Translate API"

5. Evaluation & Judging Criteria (100 Points Total)

1. **Meteorological Ingestion & Heat Index Modeling (25 Points):** Correct integration of public weather APIs and accurate computation of Heat Index/WBGT metrics.
2. **AI Vulnerability Scoring & Health Logic (25 Points):** Quality and clinical relevance of the risk stratification algorithm for vulnerable populations.
3. **Actionable Alerts & Vernacular Usability (20 Points):** Clarity, speed, and real-world utility of the advisory system across multiple languages.
4. **Dashboard UX & Geospatial Heat Mapping (15 Points):** Intuitive ward-level visual heatmaps, clear severity indicators, and mobile responsiveness.
5. **Software-Only Prototype & Code Quality (15 Points):** Clean GitHub repository, reproducible mock datasets, .env.example, and a working live deployment without requiring physical sensors.

6. Submission Guidelines

- **Code Repository:** Public/Private GitHub repository with structured commits, .env.example, sample weather cache, and complete setup instructions.
- **Live Deployment:** Working live web interface hosted on Vercel, Render, or Netlify.
- **Demo Video / Deck:** A 3-minute video showing live location search, heat-index calculation, simulated severe heatwave trigger, vulnerable group alert dispatch, and the administrative risk map.